

PROCESS DEFINITION DOCUMENT

New-Hire Benefits Enrollment

How Accrual Services HR enrolls a new hire into the Plan Year 2026 benefits programme — from the moment the candidate appears in the BenefitConnect Onboarding Queue to confirmation of their elections.

Accrual Bank · Accrual Services (Shared Services) · Benefits Administration

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Baseline date: 22 April 2026 · Anchored against BenefitConnect tenant accrual-bp-hri (Plan Year 2026 active)

Field	Value
Document Owner	Lin Kovač — Process Engineer, Accrual Services Process Engineering
Business Owner	Dana Whitfield — Director, Benefits Administration
Technical Owner	Diego Alvarez — HR Technology Lead
Sponsor	Taryn Blake — VP, Total Rewards
Classification	Internal — Accrual Services HR
Intended audience	Automation build team, HR-Tech reviewers, Internal Audit liaison
Supersedes	v1.0 (22 April 2026 — daily-batch dispatcher framing, retired)

This document describes how the Accrual Services HR Operations team enrolls each new hire into the Plan Year 2026 benefits programme. It covers the day-to-day process the HR Administrator follows in BenefitConnect, the business rules that govern the elections, the exceptions that the team handles, and

the option to lift the routine portion of the work into an automation that mirrors what the operator does in the UI. The narrative is anchored to the live BenefitConnect tenant at <https://accrual-bp-hri.azurewebsites.net>.

Document Control

Change History

Version	Date	Author	Summary of change
0.1	2026-04-08	Lin Kovač	Initial outline based on Accrual Services HR intake.
0.5	2026-04-12	Lin Kovač	As-is process walkthrough captured during ride-along with the HR-Admin operator (2h observation).
0.8	2026-04-15	Lin Kovač	Draft circulated to Benefits Administration and HR-Tech for review.
0.9	2026-04-19	Lin Kovač	Incorporated 11 SME comments; refined business rules and exception catalog.
1.0	2026-04-22	Lin Kovač	Baselined after joint walk-through with Dana Whitfield and Diego Alvarez. Daily-batch dispatcher framing.
1.1	2026-05-08	Lin Kovač	Restructured to per-employee processing anchored to the live BenefitConnect HR-Admin Onboarding Queue. Dropped CSV intake; trigger is now the queue API. Aligned API reference with the published BenefitConnect API Reference. Added Appendix C — UI Walkthrough.

Reviewers and Approvers

Role	Name	Organisation	Status
Business Owner	Dana Whitfield	Accrual Services HR — Benefits Administration	Approved 2026-05-08
Primary Operator	Maggie Lee	Accrual Services HR — HR Operations	Approved 2026-05-07
Technical Owner	Diego Alvarez	HR Technology	Approved 2026-05-08
Sponsor	Taryn Blake	Total Rewards	Approved 2026-05-08
Security Reviewer	Farah Okonkwo	IT Security — API	Approved with

Role	Name	Organisation	Status
		Governance	comment (see OI-02)
Audit Liaison	Marco Peralta	Internal Audit	Approved with comment (see OI-06)
Process Author	Lin Kovač	Accrual Services Process Engineering	Submitted

Related Documents

- Accrual Bank Benefits — Plan Year 2026 Master Plan Document (MPD-2026-Q4, issued 2025-11-01).
- BenefitConnect API Reference — public-API surface contract (vendor-published; see Appendix A).
- BenefitConnect HR-Admin Onboarding Queue — UI reference (live at /admin/onboarding; screenshot anchors in Appendix C).
- Accrual Services HR Operations Service Catalog 2026 — item S-22 “New-hire enrollment finalisation”.
- Integration Security Standard IS-2025-11 — API-key issuance, rotation, and secret storage.
- Internal Audit Memo IA-2026-04 — Controls expected over automated enrollment postings.

Open Items and Risks

ID	Item	Owner	Status	Target date
OI-01	Three legacy intake formats (post-Sutherland / post-Quorva / post-Bridgeline acquisitions) still in circulation; the partner-side onboarding bot must normalise the elections payload before invoking the consumer.	Diego Alvarez	OPEN	2026-05-15
OI-02	Dependent records are still created via the BenefitConnect UI during orientation. The /api/dependents POST/PATCH/DELETE verbs are UI-session-only per the API Reference, so the consumer must treat dependents as a precondition, not something it creates.	Diego Alvarez	OPEN	2026-07-31
OI-03	Beneficiary designations are UI-only today (PUT /api/beneficiaries is UI-session-only). Initial designation is captured by Maggie	Diego Alvarez	OPEN	2026-07-31

ID	Item	Owner	Status	Target date
	during candidate review; the consumer reads GET /api/beneficiaries for validation only.			
OI-04	Idempotency keying agreed on (EmployeeId, PlanYear) — see BR-11.	Lin Kovač	RESOLVED 2026-04-18	—
OI-05	Confirmation-email to the new hire deferred to Phase 2; Phase 1 relies on the BenefitConnect-generated confirmation notification.	Dana Whitfield	DEFERRED	Phase 2
OI-06	Internal Audit asked whether per-transaction screenshot artefacts are still required once the BenefitConnect audit log captures automated calls. Proposed: retain screenshots for the first 30 days of live running, then retire.	Marco Peralta	OPEN	2026-06-01
OI-07	The internal/partner endpoints listed in Appendix A.2 are not part of the public API surface and may return 403 API_NOT_PUBLIC under tenant-side access policies. Mitigation: rerun via the UI walkthrough in Appendix C is always available; see §9.2.	Diego Alvarez	RESOLVED 2026-05-07 — UI fallback documented	—

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1. Introduction

1.1 Purpose

This Process Definition Document (PDD) describes how the Accrual Services HR Operations team enrolls each new hire into the Plan Year 2026 benefits programme using BenefitConnect, the company's benefits-enrollment platform at <https://accrual-bp-hri.azurewebsites.net>. The document covers what the HR Administrator does on a given workday, what business rules govern the elections, what exceptions the team handles and how, and what an automation that took over the routine portion of the work would need to look like.

The work is per-employee. Each new hire who appears in the BenefitConnect Onboarding Queue is one piece of work: the operator opens that candidate's record, reviews their elections, and approves the enrollment. There is no daily batch and no end-of-day cut-off — the queue is worked continuously through the business day. The automation specification in §4–§9 is structured the same way: one transaction per new hire, triggered when the candidate appears in the queue.

1.2 Scope

In scope

- Detection of a new pending onboarding request for a single employee (queue API poll or webhook).
- Validation of the candidate record against the agreed schema (see §8.1).
- Emission of one transaction item per new hire into the NewHireOnboarding queue.
- For each transaction: precondition checks, posting of each elected benefit line, creation of the pending enrollment, and promotion to Active via the BenefitConnect partner API — and resolution of the corresponding Onboarding Queue item via POST `/api/onboarding/pending/:id/approve`.
- Per-transaction logging and operational reporting (§12).

Out of scope

- Collection of the new hire's elections. The orientation intake process is upstream and is not modified by this initiative; the consumer receives a fully-formed elections payload alongside the candidate identity.
- Creation of dependent records and designation of beneficiaries. POST/PATCH/DELETE `/api/dependents` and PUT `/api/beneficiaries` are documented as UI-session-only (BenefitConnect API Reference §3); these remain HR-Admin operations performed through the BenefitConnect UI (see OI-02, OI-03).
- Qualifying Life Event (QLE) processing — separate scenario SCN-11.
- Annual Open Enrollment batch posting — separate scenario SCN-08.
- Payroll deduction feed — handled downstream by the existing Payroll Integration (PI-2024-02).

- Carrier eligibility files — BenefitConnect emits these on its own schedule, independent of this process.

1.3 Audience

The primary audience is the automation build team responsible for implementing and operating the solution. Secondary audiences include Benefits Administration SMEs, HR Technology architects, Internal Audit reviewers, and the BenefitConnect platform team. Readers are assumed to be familiar with the Accrual Bank benefits programme at a conceptual level; plan-year specific numbers are either cited inline or deferred to the Plan Year 2026 Master Plan Document.

1.4 Conventions and Terminology

Term	Definition
Onboarding Queue	The list of pending new-hire onboarding requests visible in the BenefitConnect HR-Admin UI at /admin/onboarding. Backed by GET /api/onboarding/pending (partner-callable, public-keyed) and GET /api/admin/onboarding-queue (UI-session-only).
Candidate	An employee record in the Onboarding Queue. Each candidate carries identity, employment, and onboarding-status fields; see §8.1.
Transaction item	A single new-hire onboarding task enqueued by the dispatcher for the consumer to process. Schema in §8.2. One transaction = one new hire.
Enrollment line	A single benefit election (e.g., medical, dental, FSA health). A new-hire posting typically comprises 7–11 enrollment lines.
Draft	The server-side accumulation of enrollment lines for a given employee, materialised by successive POST /api/plans/select calls and finalised by POST /api/enroll.
Confirmation	The Active enrollment record returned by POST /api/confirm (or by POST /api/onboarding/pending/:id/approve when a Pending Confirmation exists) after the draft is promoted.
Plan Year	A 12-month benefits period beginning 1 January and ending 31 December.
Partner API-key	The X-API-Key secret issued by BenefitConnect to a partner caller. See Integration Security Standard IS-2025-11 §3 for rotation policy.
X-Employee-Id	Header (or ?employeeid= query param) that scopes a partner call to a specific employee in the tenant. Required for per-employee

Term	Definition
	operations on the partner API.

2. Process Overview

2.1 Business Context

Accrual Bank and its US subsidiaries (Accrual Capital, Accrual Services, and Accrual Trust) hire between 140 and 220 employees per month across the group's US-domiciled legal entities and a smaller cohort across the APAC and EMEA branches. Benefits eligibility begins on the first day of employment for US hires and on a region-specific equivalent for international hires; in either case a new hire's elections must be reflected in BenefitConnect as soon as practicable so that carrier eligibility files, ID-card dispatch, and payroll deductions can fall into their normal cadences.

Elections are collected during a 90-minute benefits-orientation session held within the first three business days of employment. The orientation coach captures elections on a structured intake form and records dependents and beneficiaries directly in BenefitConnect. The HR-Admin operator (today: Maggie Lee) then logs into BenefitConnect, picks up the candidate's record from the Onboarding Queue at /admin/onboarding, walks the BenefitConnect 6-step enrollment wizard on behalf of the new hire, and clicks Approve on the candidate's detail page (/admin/onboarding/:employeeId) to mark the request resolved and confirm the enrollment to Active. This per-hire UI work is what the present initiative automates.

2.2 Business Benefit

- **Reclaim HR-Admin capacity.**

At a manual throughput of ~12 minutes per hire and ~8 hires per day on a normal week, the per-employee onboarding finalisation consumes ~0.2 FTE during a normal month and escalates during hiring peaks (January, September, post-acquisition ramps). Removing the rekey frees Maggie for exception handling and new-hire coaching.

- **Reduce posting latency.**

Today, new hires commonly do not see their elections reflected in the self-service portal until 24–48 hours after orientation, which generates inbound queries to the HR-Helpline. The per-event automation completes a transaction within minutes of the candidate appearing in the queue.

- **Strengthen audit posture.**

Every automated posting carries an immutable source attribution (BR-12) and is covered by the BenefitConnect platform's own audit log.

- **Reduce IRS-limit breaches and tier inconsistencies.**

The deterministic checks in §7 are applied uniformly by the consumer; historically, two material breaches have been identified at year-end reconciliation (plan years 2023 and 2024), both traced to keystroke errors in the wizard.

2.3 Stakeholders

Role	Name	Relationship to process
Process Owner (Business)	Dana Whitfield, Director of Benefits Administration	Accountable for the end-to-end process, its controls, and its KPIs. Sign-off on business rules and exception resolution.
Sponsor	Taryn Blake, VP Total Rewards	Funds and prioritises the automation; owns the Benefits function's service commitments.
Primary Operator (as-is)	Maggie Lee, HR Administrator, Accrual Services	Logs into BenefitConnect, processes each pending candidate from the Onboarding Queue, walks the wizard, clicks Approve.
Backup Operator	Elena Morris, Senior Benefits Specialist	Handles exceptions routed out of the automation and absence-coverage during peak weeks.
Technical Owner	Diego Alvarez, HR Technology Lead	Owns BenefitConnect configuration, partner-integration secrets, and the queue infrastructure.
Platform Engineering Lead	Rohan Deshmukh, Head of Platform Engineering	Line manager for HR Technology; accountable for platform reliability and the escalation path on sustained application exceptions.
CIO (informed)	Lin Xu, Chief Information Officer	Executive accountability for IT integrations; informed on year-end controls reporting and material incidents.
API Governance	Farah Okonkwo, IT Security	Issues and rotates partner API-keys; reviews secret-handling in the automation.
Audit Liaison	Marco Peralta, Internal Audit	Reviews controls; sign-off on retention of evidence and the automated audit trail.
Platform Vendor Contact	BenefitConnect Platform Team (vendor-side)	Maintains the published API surface (see Appendix A) and responds to partner integration tickets.

2.4 Personas

The following persona sketches are referenced throughout the as-is narrative and the exception routing.

- **Dana Whitfield** — Director of Benefits Administration, 14 years in the function. Measures the team on latency (intake-to-active), posting accuracy, and year-end reconciliation exceptions. Conservative on process changes that could affect audit posture.
- **Maggie Lee** — HR Administrator, Accrual Services. Primary operator of the as-is process. Logs into BenefitConnect with admin credentials; her account in the live tenant is `maggie.lee@accrualbank.example` with role HR Administrator. She processes pending candidates one at a time from the Onboarding Queue, walking the wizard for each new hire and clicking Approve to confirm.
- **Elena Morris** — Senior Benefits Specialist, cross-trained on all plan lines, three years at Accrual Bank. Becomes the primary resolver of exceptions routed from the consumer.
- **Diego Alvarez** — HR Technology Lead. Owns the BenefitConnect tenant configuration, the API-key inventory, and the automation platform. Receives consumer-level alerts and owns the resumption protocol.
- **Leilani Tan (worked example)** — Associate, Commercial Lending — APAC; Accrual Capital division; based at Accrual Bank — Singapore HQ; manager James Okafor. Employee number ACR-00894, BenefitConnect ID E1004, hire date 2026-04-26. Used as the worked example throughout the document; her live-tenant record at `/admin/onboarding/E1004` is the canonical reference for the candidate schema (§8.1) and the worked trace (Appendix B).

3. As-Is Process

3.1 Trigger and Cadence

The as-is process is triggered per-new-hire by the appearance of a pending candidate in the BenefitConnect Onboarding Queue. Maggie checks the queue several times per business day — typically at 09:30, 13:00, and 16:00 — and works the queue most-urgent-first (the UI sorts by Days-Since-Hire descending). There is no daily batch run; each candidate is its own unit of work.

3.2 End-to-End Narrative (UI walkthrough)

The following narrative reflects the per-employee process observed during the on-site walk-through with Maggie Lee. It is anchored to specific URLs and UI elements in the live BenefitConnect tenant; see Appendix C for a step-by-step UI reference. Step numbers are indicative of the order Maggie follows.

1. Maggie navigates to <https://accrual-bp-hri.azurewebsites.net/login> and signs in with her HR Administrator credentials. She lands on the HR-Admin shell whose left sidebar shows: HR Operations (Dashboard, Onboarding Queue, Onboard New Hire) and My Account (My Profile, Documents).
2. She clicks Onboarding Queue (URL: </admin/onboarding>). The page renders a single table titled “Pending HR Onboarding Requests — New hires awaiting benefits enrollment · sorted most-urgent-first” with columns Employee, Department, Hire Date, Days Since Hire, Deadline, and Status. Status values seen in production today include Not Started, Draft Saved, Pending Confirmation, and Overdue.
3. Maggie selects the candidate she will work on (for the worked example: Leilani Tan · E1004 · Commercial Lending — APAC · Hire 2026-04-26 · Status Not Started · Deadline 2026-05-26). She clicks View on the row, which opens the candidate detail page at </admin/onboarding/E1004>.
4. On the candidate detail page she verifies the two information panels: Candidate (Preferred Name, Email, Phone, Date of Birth, Employee Number ACR-00894) and Employment (Job Title, Department, Division, Manager, Work Location, Employee Type, Start Date). She cross-references these against the signed intake form (Form HR-014, paper during orientation, scanned to SharePoint after); name, hire date, and legal entity must match.
5. If the candidate is in Not Started, Maggie navigates to the BenefitConnect enrollment wizard for that employee and walks the 6 steps observed in the live UI: (1) Personal Info — read-only, sourced from HRIS; (2) Dependents — confirms what was captured during orientation; (3) Medical; (4) Dental / Vision / Life; (5) Savings & 401(k); (6) Review & Submit. For each elected line she selects the plan and tier from the catalogue dropdown, enters any numeric amounts, and lets the wizard compute the per-paycheck premium. She spot-checks computed premiums against the intake-form values.

6. On Review & Submit she visually confirms each line and clicks Submit. The candidate's onboarding status transitions from Not Started → Draft Saved → Pending Confirmation.
7. Maggie returns to /admin/onboarding/{employeeId} (the candidate detail page) and clicks the green Approve button. The page displays the explanation: “Approving will mark this request resolved and (if a pending enrollment exists) confirm it.” The candidate's status transitions to Active and the request is removed from the pending queue.
8. She logs the posting in the shared tracking workbook (BenefitsPostingLog.xlsx), columns: EmployeeId, HireDate, ProcessedAt, Outcome, Notes.
9. Exceptions (intake-form / HRIS mismatch, tier/dependent inconsistency, IRS limit breach, voluntary-life EOI requirement) are flagged in-line in the tracking workbook with an Exception-Code column and emailed to Elena Morris for same-day resolution.
10. At the end of the day Maggie emails Dana a one-line summary: total candidates processed, total exceptions, and any stragglers carried over.

3.3 Pain Points Identified

- The per-hire median duration is 12 minutes of wall-clock keystroke time, of which ~75% is repetitive data entry and ~25% is the mandatory spot-checks.
- Age-banded voluntary-life premium is displayed only after tier/amount entry and is the most common source of keystroke error — a mistyped digit changes the payroll deduction for 26 pay periods.
- The wizard submits elections as a single transaction — a validation failure on the final step requires rework of the earlier steps.
- Evidence retention relies on screenshots, which Internal Audit has flagged as brittle (OI-06).
- During hiring peaks the queue depth exceeds sustainable manual throughput and candidates slip past their enrollment deadline (Status: Overdue), generating HR-Helpline tickets and downstream payroll/eligibility exceptions.

3.4 As-Is Metrics

Metric	Value	Source
Manual median duration per candidate	12 min (p50), 19 min (p90)	Maggie walk-through + 2 weeks of BenefitsPostingLog sampling
Daily intake volume (normal month)	6–10 new hires/day	Onboarding Queue arrivals, 2025 calendar year
Daily intake volume (peak)	22–40 new hires/day	Onboarding Queue arrivals, Jan 2025 + Sep 2025
Elections per hire	7–11 lines (mode 9)	BenefitsPostingLog.xlsx, 2025

Metric	Value	Source
Latency (queue-arrival → Active) — normal	4–8 business hours (same-day)	BenefitsPostingLog.xlsx
Latency — peak	24–48 business hours	BenefitsPostingLog.xlsx; HR-Helpline ticket correlations
Exception rate	~4.2% of postings	BenefitsPostingLog.xlsx, 2025
Audit-relevant rekey defects (2023 + 2024)	2 material (both IRS-limit overages)	Year-end reconciliation

4. To-Be Process

4.1 Guiding Principles

- The elections-collection activity (orientation session, intake form, dependent creation in BenefitConnect) is unchanged. Automation begins after the candidate appears in the Onboarding Queue.
- The unit of work is one new hire. The process decomposes cleanly into a dispatcher that listens for queue arrivals and emits one transaction item per candidate, and a consumer that processes one transaction at a time. The two are decoupled by a durable queue.
- Every automated posting is idempotent on (EmployeeId, PlanYear) so that reruns are safe by construction (BR-11).
- Exceptions are first-class. Every transaction terminates in either Success, Business-Exception (routed to Elena), or Application-Exception (retried, then escalated to Diego). There is no silent failure mode.
- The automation mirrors what the HR-Admin operator does in the UI; the UI walkthrough in §3.2 / Appendix C is always available as a fallback for any transaction the automation declines or escalates.

4.2 High-Level Flow

The automation runs as two cooperating components:

11. A dispatcher polls the BenefitConnect Onboarding Queue at a configurable cadence (default every 5 minutes during business hours) by calling GET /api/onboarding/pending with the partner X-API-Key. For each newly-arrived candidate it emits one transaction item into the NewHireOnboarding queue and writes a dispatcher-pickup log row. (Optional Phase 2: webhook-driven dispatch if BenefitConnect publishes one.)
12. A consumer drains the queue, processing one transaction at a time. For each item it sets X-Employee-Id to the candidate's id, applies the preconditions in §7, posts each enrollment line via POST /api/plans/select, materialises a pending enrollment via POST /api/enroll, and confirms it to Active either via POST /api/confirm or via POST /api/onboarding/pending/:id/approve (which both confirms the pending enrollment and resolves the queue item in one call). Outcomes are logged per transaction.

4.3 Target Metrics

Metric	Target	Notes
Dispatcher poll cycle	≤ 30 s per cycle (no candidates) / ≤ 60 s with candidates	GET /api/onboarding/pending plus enqueue.
Consumer duration per transaction	≤ 45 s (p50), ≤ 120 s (p95)	Wall-clock, including all API calls.

Metric	Target	Notes
Run success rate	≥ 98% transactions Active without human touch	Measured weekly.
Queue-arrival to Active latency	≤ 30 min during business hours	Inclusive of dispatcher poll wait.
Business-exception resolution SLA	Same business day (handled by Elena)	Escalation path in §10.1.
Application-exception escalation SLA	Within 30 minutes of third retry	Paged to HR-Tech on-call.
Audit-evidence completeness	100% of Active postings carry source = api:accrual-onboarding-bot-v2	See BR-12.

5. Process Architecture

5.1 Component View

The solution is a two-component dispatcher/consumer topology, anchored to the BenefitConnect Onboarding Queue:

- **Dispatcher** — scheduled, short-lived, stateless. Polls GET /api/onboarding/pending; emits one transaction per newly-arrived candidate. See §5.2.
- **Transaction queue** — durable, FIFO within priority tier, at-least-once delivery. See §5.3.
- **Consumer** — triggered by queue-not-empty. Processes one transaction at a time; idempotent. See §5.4.
- **Shared configuration store** — holds the X-API-Key secret, the BenefitConnect base URL, queue name, poll cadence, and SLAs.

5.2 Dispatcher

Aspect	Specification
Trigger	Time-based, every 5 minutes during business hours (Mon–Fri 07:00–19:00 local). Off-hours cadence is configurable; default 30 minutes.
Pre-check	GET /api/public/health and GET /api/public/status before each poll. Abort the cycle with a structured warning on failure (see §10.2).
Source	GET /api/onboarding/pending called with X-API-Key. Returns the list of candidates with status in {Not Started, Draft Saved, Pending Confirmation}.
Validation	For each candidate: required identity fields present (id, username, hireDate, planYear), elected plan codes resolve via GET /api/public/plans, BR-01 (hire-date window) holds.
De-duplication	If a transaction already exists for (EmployeeId, PlanYear) in-flight or completed within the look-back window, skip the candidate.
Output	One transaction item per enqueue-ready candidate, emitted to queue NewHireOnboarding (§5.3), annotated with DispatcherPollId.
On invalid	Candidates that fail validation are not enqueued; an exception row is written and Elena is emailed for triage.
On API error	5xx → 3 retries with exponential backoff (5s/20s/60s); 401 → one secret re-read then escalate; 403 API_NOT_PUBLIC → escalate to Diego (treat as platform-side configuration change).

5.3 Transaction Queue

Aspect	Specification
Queue name	NewHireOnboarding
Delivery semantics	At-least-once (consumer guarantees idempotency per BR-11).
Ordering	FIFO within priority. Priority defaults to Normal. Overdue candidates may be promoted to High.
Item expiry	24 hours. An item not processed within 24 hours is auto-abandoned and routed to Elena for manual handling.
Auto-retry	1 automatic retry on Application Exception, with a 5-minute backoff (see §10.2). No auto-retry on Business Exception.
Deduplication key	(EmployeeId, PlanYear). The consumer enforces the DUPLICATE_ACTIVE no-op defined in BR-11 rather than relying on queue-level dedup.
Backpressure	If the queue depth exceeds 120 (roughly 3× a peak-day cohort), dispatching is paused and Diego is alerted.

5.4 Consumer

Aspect	Specification
Trigger	Queue-not-empty. Starts automatically; stops when the queue is drained.
Concurrency	Single-threaded in Phase 1. Multi-worker concurrency is deferred until volume demands it (not before 2026-Q4).
Authentication	X-API-Key header, loaded once per worker lifetime from the shared secret store. The key is scoped to this process by HR Technology (partner identity: accrual-onboarding-bot-v2).
Per-employee scope	Every per-employee API call carries X-Employee-Id: <candidate.id>. The header (or ?employeeId= equivalent) is mandatory for the partner API.
Per-item flow	See §6.2. Eleven steps, terminating in Success, Business Exception, or Application Exception.
Per-item logging	One row per transaction in BenefitsPostingLog.xlsx — columns: TransactionId, EmployeeId, PlanYear, StartedAt, FinishedAt, OutcomeCode, EnrollmentId, LinesPosted, ExceptionDetail.

Aspect	Specification
Run summary	Daily summary email to dl-hrops-benefits — RunId, Started, Finished, ItemsProcessed, Successes, BusinessExceptions, ApplicationExceptions.

6. Detailed Process Steps

Each step below is expressed as a unique ID, a short name, the action performed (including the inputs consumed), the outputs produced, and the exceptions that may be raised. Exception codes are catalogued in §10.

6.1 Dispatcher Steps

ID	Step	Action	Outputs	Exceptions
D-01	Confirm business hours	Use the cadence appropriate for the current time-of-day. Off-hours uses a 30-minute cycle.	— (rate-limit only)	—
D-02	Pre-check service	GET /api/public/health and GET /api/public/status. Abort cycle on failure.	Service-status snapshot	BC_UNAVAILABLE (app)
D-03	Poll Onboarding Queue	GET /api/onboarding/pending with X-API-Key. Receive list of candidates.	Candidate list	AUTH_REQUIRED / INVALID_API_KEY (app); API_NOT_PUBLIC (app)
D-04	Validate each candidate	For each candidate: required fields present, plan codes resolve via GET /api/public/plans, BR-01 (hire-date window) holds.	Enqueue-ready candidates; exceptions set	INVALID_CANDIDATE (biz, per row); BR01_HIRE_OUT_OF_WINDOW (biz)
D-05	Deduplicate	Skip candidates already in-flight or completed for (EmployeeId, PlanYear) within the look-back window.	Filtered candidate set	—
D-06	Enqueue transactions	Emit one transaction item per enqueue-ready candidate to the NewHireOnboarding queue.	Queue populated	QUEUE_UNAVAILABLE (app)
D-07	Record	Append a poll-summary row to	Poll row	— (best-

ID	Step	Action	Outputs	Exceptions
	dispatcher-poll	BenefitsPostingLog.xlsx.		effort)

6.2 Consumer Steps

ID	Step	Action	Outputs	Exceptions
C-01	Dequeue transaction	Block for the next item; fetch it and mark InProgress.	Transaction item	— (infrastructure)
C-02	Load API-key	Load the X-API-Key from the secret store; cache for the worker lifetime.	API-key in memory	SECRET_UNAVAILABLE (app)
C-03	Identity assertion	GET /api/employees/me with X-Employee-Id: <candidate.id>; assert response.id === transaction.EmployeeId.	Employee record	IDENTITY_MISMATCH (biz); AUTH_REQUIRED / INVALID_API_KEY (app); API_NOT_PUBLIC (app)
C-04	Dependent pre-check	GET /api/dependents (X-Employee-Id scoped). Assert each id in transaction.Dependents is present on the employee record.	Dependent set	DEPENDENTS_MISSING (biz); API_NOT_PUBLIC (app)
C-05	Beneficiary pre-check	GET /api/beneficiaries. Assert primary and contingent allocations each sum to 100%; a primary is required if any life line is elected.	Beneficiary set	BENEFICIARIES_MISSING (biz); BENEFICIARIES_INVALID_SUM (biz); API_NOT_PUBLIC

ID	Step	Action	Outputs	Exceptions
				(app)
C-06	Duplicate-active check	GET /api/enrollments/current. If an Active enrollment exists for PlanYear, mark DUPLICATE_ACTIVE no-op (BR-11) and jump to C-11 with OutcomeCode = SUCCESS_NO_OP.	OutcomeCode (possibly)	— (no exception on duplicate)
C-07	Post each line	For each elected line, in the order defined by BR-05, POST /api/plans/select with the payload shape defined in §8.3. Abort the transaction on any 4xx.	Server-side draft progressively built	PLAN_SELECTION_REJECTED (biz); PLAN_CATALOGUE_MISMATCH (biz); 5xx (app)
C-08	Reconcile draft	GET /api/enroll/draft. Assert the draft contains exactly the elected lines and amounts.	Reconciled draft	DRAFT_RECONCILE_MISMATCH (biz)
C-09	Create pending enrollment	POST /api/enroll. Record enrollmentId and status = Pending Confirmation.	enrollmentId	ENROLLMENT_REJECTED (biz); 5xx (app)
C-10	Confirm to Active	POST /api/onboarding/pending/<candidate.id>/approve — confirms the pending enrollment and resolves the queue item in one call. (Alternative: POST /api/confirm followed by a separate approve.)	confirmationNumber; Active record; queue resolved	CONFIRM_REJECTED (biz); 5xx (app)
C-11	Log outcome	Append a transaction row to BenefitsPostingLog.xlsx; mark the local queue item Successful, BusinessException, or ApplicationException as appropriate.	Log row; queue item resolved	— (best-effort)

7. Business Rules

Every rule below is an assertion evaluated by the consumer (unless marked D for Dispatcher) and carries an exception code raised when violated. Rules are numbered for traceability; the numbering is stable across document revisions.

ID	Rule	Where enforced	Exception on violation
BR-01	Hire date must fall within the previous 30 calendar days. Older hires are out-of-scope; route to HR for special handling.	D (D-04)	BR01_HIRE_OUT_OF_WINDOW
BR-02	All elected lines must specify the plan year matching the hire date. Hires between 15 November and 31 December may carry two plan years (current and next); in that case each line must name its plan year explicitly.	D + C	BR02_PLAN_YEAR_MISMATCH
BR-03	If the medical tier is EE+Spouse or Family, at least one dependent of relationship Spouse must be on file. If the tier is EE+Child(ren) or Family, at least one dependent of relationship Child (or Stepchild) must be on file.	C (C-04)	BR03_TIER_INCONSISTENT
BR-04	Voluntary-life face amount must not exceed the lower of 5× annual salary or \$500,000. Any amount above \$250,000 requires Evidence of Insurability (EOI) which is not handled by this process — reject.	C (C-07)	BR04_VOL_LIFE_EXCEEDS_GI_OR_CAP
BR-05	Plan-type posting order: medical → dental → vision → basic life → voluntary life → AD&D → STD → LTD → FSA-health → FSA-dependent-care → HSA → commuter-transit → commuter-parking → 401(k). This order matches the orientation-wizard order and preserves audit parity with the manual process.	C (C-07)	— (order-only)
BR-06	IRS 2026 contribution limits: Healthcare FSA \$3,300 annual; Dependent-care FSA \$5,000 (single / MFJ) or \$2,500 (MFS); HSA self-only \$4,300 / family \$8,550; Commuter transit \$325/mo; Commuter parking \$325/mo; 401(k) employee deferral \$23,500; 50+ catch-up \$7,500. The consumer rejects any election above these ceilings.	C (C-07)	BR06_IRS_LIMIT_EXCEEDED

ID	Rule	Where enforced	Exception on violation
BR-07	HSA contribution > 0 is valid only when the elected medical plan is an HDHP (plan type hdhp in the catalogue).	C (C-07)	BR07_HSA_NON_HDHP
BR-08	Dependent-care FSA contribution > 0 requires at least one dependent under age 13 (as of 31 December of the plan year) OR a disabled-dependent flag.	C (C-07)	BR08_DCFSA_NO_ELIGIBLE_DEPENDENT
BR-09	The consumer does not independently compute any premium or match; it consumes the values returned by /api/plans/select and /api/enroll. Discrepancies between intake-form expected premium and server-computed premium greater than 5% are flagged for review but do not block posting.	C (C-07, C-08)	BR09_PREMIUM_VARIANCE (warning, non-blocking)
BR-10	A transaction is considered complete only after the candidate's onboarding-queue item is resolved with status = Active (via POST /api/onboarding/pending/:id/approve or equivalent).	C (C-10)	BR10_CONFIRM_FAILURE
BR-11	Idempotency: if an Active enrollment exists for (EmployeeId, PlanYear) before posting begins, the transaction is marked DUPLICATE_ACTIVE with OutcomeCode = SUCCESS_NO_OP and no further mutating API calls are made.	C (C-06)	— (no-op)
BR-12	Every Active enrollment created by this process must carry source = api:accrual-onboarding-bot-v2 in the BenefitConnect audit record. This is set server-side by virtue of the partner API-key and must not be overridden.	Platform (server-side)	— (post-condition, audited)
BR-13	The consumer must not modify dependents or beneficiaries (POST/PATCH/DELETE /api/dependents and PUT /api/beneficiaries are documented UI-session-only). If a precondition check (C-04 / C-05) fails, the transaction is routed as a business exception.	C (C-04, C-05)	DEPENDENTS_MISSING / BENEFICIARIES_MISSING

8. Input and Output Specifications

8.1 Candidate Schema (Onboarding Queue record)

Each candidate is a record returned by GET /api/onboarding/pending and rendered in the HR-Admin UI at /admin/onboarding/:employeeId. The fields below are the live-tenant shape (canonical reference: GET /api/onboarding/pending/E1004 — Leilani Tan). The consumer reads identity and employment fields directly; the elections payload is supplied separately by the upstream orientation/intake source.

Field	Type	Req	Example	Notes
id	string	Y	E1004	BenefitConnect employee id; primary key for X-Employee-Id.
legalName	string	Y	Leilani Tan	Used for human-readability in the log only.
preferredName	string	N	Leilani	Defaults to first name.
email	string	Y	leilani.tan@accrualbank.example	Tenant email.
phone	string	N	+65 8123 5502	
dateOfBirth	date (ISO-8601)	Y	1996-09-24	
employeeNumber	string	Y	ACR-00894	HRIS payroll number (separate from BenefitConnect id).
jobTitle	string	Y	Associate, Commercial Lending	
department	string	Y	Commercial Lending — APAC	
division	enum {Accrual Services, Accrual Capital, Accrual Trust}	Y	Accrual Capital	

Field	Type	Req	Example	Notes
manager	string	Y	James Okafor	
workLocation	string	Y	Accrual Bank — Singapore HQ	
employeeType	enum {Full-Time, Part-Time, Contractor, Intern}	Y	Full-Time	
startDate	date (ISO-8601)	Y	2026-04-26	Hire date.
status	enum {Not Started, Draft Saved, Pending Confirmation, Active, Overdue}	Y	Not Started	Onboarding workflow status.
enrollmentDeadline	date (ISO-8601)	Y	2026-05-26	30 days from hire.

8.2 Transaction Item Schema

Each dispatcher emits one transaction per enqueue-ready candidate. The transaction item is a JSON document with the following shape (wire format depends on the queue platform; logical shape is what matters).

```
{
  "TransactionId": "TXN-20260510-0007",
  "DispatcherPollId": "POLL-20260510-1130",
  "EmployeeId": "E1004",
  "Username": "leilani",
  "HireDate": "2026-04-26",
  "PlanYear": 2026,
  "IntakeFormRef": "HR014-2026-03928",
  "Elections": {
    "medical": { "planId": "PLN-MED-PPO", "tier": "EE" },
    "dental": { "planId": "PLN-DEN-PPO", "tier": "EE" },
    "vision": { "planId": "PLN-VIS-STD", "tier": "EE" },
    "basicLife": { "planId": "PLN-LIFE-BASIC", "enrolled": true },
    "voluntaryLife": { "planId": "PLN-LIFE-VOL", "amount": 100000 },
    "std": { "planId": "PLN-STD", "enrolled": true },
    "ltd": { "planId": "PLN-LTD", "enrolled": true },
    "fsaHealth": { "planId": "PLN-FSA-HC", "annualContribution": 2200 },
    "commuter": { "planId": "PLN-COMMUTE", "transitMonthly": 150, "parkingMonthly": 0 },
    "retirement": { "planId": "PLN-401K", "contributionPct": 8, "rothPct": 0 }
  },
}
```

```

"Dependents": [],
"BeneficiariesOnFile": true,
"DiscoveredAt": "2026-05-10T11:30:14+08:00",
"EnqueuedAt": "2026-05-10T11:30:16+08:00"
}

```

8.3 POST /api/plans/select — Payload Shapes by Plan Type

The shape of the body posted to /api/plans/select varies by plan type. The following matrix is the contract used by the consumer; refer to Appendix A.3 for the endpoint metadata.

Plan type	Body
medical / dental / vision	{ planId, tier }
basicLife	{ planId, enrolled }
voluntaryLife	{ planId, amount }
std / ltd	{ planId, enrolled }
fsaHealth / fsaDependent	{ planId, annualContribution }
hsa	{ planId, annualContribution } (requires HDHP medical — BR-07)
commuter	{ planId, transitMonthly, parkingMonthly }
retirement (401(k))	{ planId, contributionPct, rothPct }

8.4 Outputs

- **Per-transaction:** a log row in BenefitsPostingLog.xlsx as specified in §5.4.
- **Per-day:** a daily summary email to dl-hrops-benefits.
- **Platform-side:** the Active enrollment record in BenefitConnect (carrying source = api:accrual-onboarding-bot-v2 per BR-12) and the derived entries in the BenefitConnect audit log.

9. System Interfaces

9.1 Systems in Play

System	Role	Owner
BenefitConnect tenant accrual-bp-hri	Source-of-truth for the Onboarding Queue and the system of record for enrollments.	HR Technology (Diego) + vendor platform team
Orientation/Intake source	Upstream — supplies the elections payload paired with each candidate.	HR Technology (Diego)
Transaction queue platform	Durable queue for NewHireOnboarding.	Automation Platform Team
Secret store	Holds the X-API-Key for BenefitConnect.	IT Security (Farah)
BenefitsPostingLog.xlsx	Human-readable audit log maintained on SharePoint.	Benefits Administration (Dana)
Email distribution	dl-hrops-benefits for daily summaries; dl-hrtech-oncall for escalations.	Accrual Services HR-Ops

9.2 BenefitConnect API — Integration Model

BenefitConnect exposes three tiers of HTTP surface that this process touches. The vendor's published API Reference is the contractual source-of-truth; Appendix A reproduces the relevant subset.

9.2.1 Public anonymous (no key required)

Always available. Used by the dispatcher for service liveness and plan-code validation.

Method	Path	Used by	Purpose in this process
GET	/api/public/health	Dispatcher (D-02)	Liveness check before each poll cycle.
GET	/api/public/status	Dispatcher (D-02)	Asserts current planYear and openEnrollment window.
GET	/api/public/version	Dispatcher (start-up)	Logged in run summary.
GET	/api/public/plans	Dispatcher (D-04)	Plan catalogue; validates plan codes in the elections payload.

Method	Path	Used by	Purpose in this process
GET	/api/public/plans/:id	Dispatcher (D-04)	Point lookups.
GET	/api/public/plan-types	Dispatcher (D-04)	Enum of plan types.
GET	/api/public/carriers	— (reference only)	Used during document preparation.

9.2.2 Public, X-API-Key required (documented & guaranteed-stable)

Per the BenefitConnect API Reference, only two endpoints are part of the documented partner-callable surface: GET /api/employees/me and GET /api/documents. Both require X-API-Key and accept the X-Employee-Id header (or ?employeeid= query param) for per-employee scoping. These two are the only partner endpoints with a versioned contract from the vendor.

Method	Path	Used by	Purpose
GET	/api/employees/me	Consumer (C-03)	Identity assertion (X-Employee-Id scoped).
GET	/api/documents	Consumer (post-confirm)	Optional: fetch the confirmation statement for the audit pack.

9.2.3 Internal endpoints (back the BenefitConnect web app; partner-callable as a UI shortcut)

The endpoints below are not part of the published public-API surface. They are the same HTTP calls the BenefitConnect web app issues from the browser as Maggie walks the wizard and approves the candidate — i.e. they are a programmatic shortcut around the UI rather than an exclusive capability set. The vendor's API Reference describes them as “subject to tenant-side access policies” and notes that they may return 403 with code: API_NOT_PUBLIC at any time. Production integrations should not depend on them.

Important framing: every operation these endpoints perform — adding a plan/tier/contribution to the draft, building the draft, creating the pending enrollment, confirming to Active, approving the queue item — is achievable today by Maggie (or by UI automation) walking the BenefitConnect UI flow described in §3.2 and Appendix C. The choice to call them as APIs is purely an efficiency, reliability, and audit-traceability decision; if the internal surface ever returns 403 API_NOT_PUBLIC, the same outcomes can be produced by reverting to the UI flow.

Method	Path	Used by	Purpose
GET	/api/dependents	Consumer (C-04)	Dependent pre-check (read-only) (OI-02).
GET	/api/beneficiaries	Consumer (C-05)	Beneficiary pre-check

Method	Path	Used by	Purpose
			(read-only) (OI-03).
GET	/api/enrollments/current	Consumer (C-06)	Idempotency check (BR-11).
POST	/api/plans/select	Consumer (C-07)	Per-line election; payload shape in §8.3.
GET	/api/enroll/draft	Consumer (C-08)	Draft reconciliation before commit.
POST	/api/enroll	Consumer (C-09)	Create pending enrollment from draft.
POST	/api/confirm	Consumer (alternative C-10)	Promote pending to Active (without resolving the queue item).
GET	/api/onboarding/pending	Dispatcher (D-03)	List pending onboarding candidates (cross-employee).
GET	/api/onboarding/pending/:id	Consumer (pre-C-03)	Per-candidate detail.
POST	/api/onboarding/pending/:id/approve	Consumer (C-10, preferred)	Confirm pending enrollment AND resolve the queue item in a single call.

Because this set of endpoints is not under a published public-API contract, any 403 API_NOT_PUBLIC or unexpected response shape should be treated as a tenant-side configuration change, escalated to Diego, and worked around via the UI flow (Appendix C) until the platform contact has clarified.

9.3 Authentication and Secret Handling

- The X-API-Key secret is provisioned by IT Security (Farah) per Integration Security Standard IS-2025-11 §3.
- Scope: this key is allocated exclusively to the accrual-onboarding-bot-v2 partner identity; it is not reused by any other integration.
- Rotation: every 180 days or on role-change of the human custodian, whichever is sooner. The consumer re-reads the secret on a 401.

- Per-employee scoping: every per-employee call includes X-Employee-Id: <candidate.id>. Omitting the header causes the call to fall back to the API-key's default employee, which is undesired for this process.
- Failure mode: on 401 the consumer attempts one secret re-read before escalating as INVALID_API_KEY (§10.2). On 403 with code = API_NOT_PUBLIC the consumer escalates to HR-Tech on-call and, if a candidate is mid-flight, recommends the operator complete the candidate via the UI walkthrough (§3.2 / Appendix C).

10. Exception Handling

Exceptions are partitioned into Business Exceptions (attributable to the data or the state of the employee record; resolved by Benefits Administration) and Application Exceptions (attributable to the automation or its dependencies; resolved by HR Technology). Every exception is logged with its transaction context and its resolution path is defined below.

10.1 Business Exceptions

Code	Trigger	Resolution path	SLA
BR01_HIRE_OUT_OF_WINDOW	Hire date older than 30 days at dispatch time (BR-01).	Dispatcher keeps the candidate out of the queue and emails Maggie. Maggie confirms whether to run as a special case or defer.	Same day
BR02_PLAN_YEAR_MISMATCH	One or more elected lines do not match the plan year implied by the hire date (BR-02).	Transaction routed to Elena; intake-form re-validation required.	Same day
BR03_TIER_INCONSISTENT	Medical tier names a spouse or children dependents absent from the employee record (BR-03).	Routed to Elena; likely a dependent-entry gap from orientation (OI-02).	Same day
BR04_VOL_LIFE_EXCEEDS_GI_OR_CAP	Voluntary life face amount exceeds guaranteed-issue limit or absolute cap (BR-04).	Routed to Elena; EOI workflow is out-of-scope for this automation.	Same day
BR06_IRS_LIMIT_EXCEEDED	Any contribution above the IRS limit in effect for the plan year (BR-06).	Routed to Elena; intake-form correction or election reduction.	Same day
BR07_HSA_NON_HDHP	HSA contribution > 0 while the elected medical plan is not an HDHP (BR-07).	Routed to Elena.	Same day
BR08_DCFSA_NO_ELIGIBLE_DEPENDENT	Dependent-care FSA elected without an eligible dependent (BR-08).	Routed to Elena.	Same day
DEPENDENTS_MISSING	One or more dependents referenced in the transaction are not present on the employee record (C-04).	Routed to Elena; dependent creation by Maggie via the BenefitConnect UI, then rerun.	Same day

Code	Trigger	Resolution path	SLA
BENEFICIARIES_MISSING	A life line is elected but no primary beneficiary is on file (C-05).	Routed to Elena; beneficiary capture by Maggie via the BenefitConnect UI, then rerun.	Same day
BENEFICIARIES_INVALID_SUM	Primary or contingent beneficiary allocations do not sum to 100% (C-05).	Routed to Elena; correction by Maggie, then rerun.	Same day
PLAN_SELECTION_REJECTED	POST /api/plans/select returned a 4xx with a validation message (C-07).	Routed to Elena; the validation message dictates the correction.	Same day
PLAN_CATALOGUE_MISMATCH	Plan id in elections payload is not present in the catalogue at run time (D-04 / C-07).	Routed to Diego; most likely a plan-code retirement or intake-tool drift.	Next business day
DRAFT_RECONCILE_MISMATCH	The server-side draft does not match the transaction's elections (C-08).	Routed to Diego for triage.	Next business day
ENROLLMENT_REJECTED / CONFIRM_REJECTED	POST /api/enroll, /api/confirm, or /api/onboarding/pending/:id/approve returned a 4xx (C-09 / C-10).	Routed to Elena; interpret the error message in the API response.	Same day
IDENTITY_MISMATCH	GET /api/employees/me returns a different EmployeeId than the transaction (C-03).	Routed to Diego; likely a header-scope (X-Employee-Id) bug.	Immediate (blocks worker)

10.2 Application Exceptions

Code	Trigger	Retry policy	Escalation
BC_UNAVAILABLE	GET /api/public/health or /api/public/status fails on poll start.	Dispatcher skips the cycle.	If 3 consecutive cycles fail: page HR-Tech on-call.

Code	Trigger	Retry policy	Escalation
QUEUE_UNAVAILABLE	Local queue platform is not reachable.	3 attempts with 2-minute backoff.	Page HR-Tech on-call.
SECRET_UNAVAILABLE	Consumer C-02 — secret store is not reachable.	2 attempts with 1-minute backoff.	Page IT Security on-call.
AUTH_REQUIRED	Any endpoint returns 401 with no key supplied (shouldn't happen).	No retry.	Page HR-Tech on-call.
INVALID_API_KEY	Any endpoint returns 401 with code INVALID_API_KEY.	One secret re-read, then no retry.	Page IT Security on-call.
API_NOT_PUBLIC	Any internal endpoint returns 403 with code API_NOT_PUBLIC (tenant-side configuration).	No retry.	Page HR-Tech on-call. Recommend operator complete the in-flight candidate via the UI flow (§3.2 / Appendix C).
HTTP_5XX	Any endpoint returns a 5xx.	3 attempts with exponential backoff (5s/20s/60s).	After the third failure, escalate to HR-Tech on-call and fail the transaction.
NETWORK_TIMEOUT	TCP timeout or TLS handshake failure.	Same as HTTP_5XX.	Same as HTTP_5XX.

10.3 Escalation Summary

A single table, suitable for the operational runbook:

Who sees what	Business Exceptions	Application Exceptions
First responder	Elena Morris (Senior Benefits Specialist)	HR-Tech on-call (Diego's rota)
Notification	Per-transaction email to Elena; daily digest to Dana.	Immediate page to on-call; daily summary to dl-hrtech-oncall.
Record-keeping	Row in BenefitsPostingLog.xlsx with OutcomeCode and ExceptionDetail.	Same row, plus incident ticket in the HR-Tech queue.
Reprocessing	Re-queue the transaction via the Rerun utility after the data is fixed.	Automatic upon retry; manual after escalation; UI fallback per Appendix C.

11. Volumetrics and Service Levels

11.1 Volume

Window	Daily new-hire arrivals	Elections/hire	Peak-hour consumer load
Normal weekday (Feb–Jul, Oct–Nov)	6–10 hires/day	7–11 lines	6–10 transactions across the working day
Peak weekday (Jan, early-Sep)	22–40 hires/day	7–11 lines	Up to 40 transactions; bursts at orientation-completion times
Post-acquisition weekday	20–60 hires/day	7–11 lines	Consumer may run continuously through business hours
Weekend / public holiday	Rare	—	Dispatcher off-hours cadence

11.2 Service Levels

SLA	Commitment
Dispatcher poll cycle	≤ 60 seconds wall-clock per cycle (with candidates).
Consumer per-transaction (p50)	≤ 45 seconds wall-clock.
Consumer per-transaction (p95)	≤ 120 seconds wall-clock.
Queue-arrival to Active (p95)	≤ 30 minutes during business hours.
Business-exception triage (Elena)	Same business day.
Application-exception escalation (HR-Tech)	Within 30 minutes of third retry.
Daily summary email	By 19:30 local each business day.
Evidence retention (BenefitsPostingLog.xlsx)	7 years.
Evidence retention (BenefitConnect audit log)	Per vendor contract — 10 years.

12. Reporting and Logging

12.1 Per-Transaction Logging

A single row is appended to BenefitsPostingLog.xlsx per transaction, irrespective of outcome.

Required columns:

- TransactionId, DispatcherPollId, EmployeeId, Username, HireDate, PlanYear.
- StartedAt, FinishedAt (ISO-8601, local timezone).
- OutcomeCode ∈ { SUCCESS, SUCCESS_NO_OP, BUSINESS_EXCEPTION, APPLICATION_EXCEPTION }.
- EnrollmentId (on success); ConfirmationNumber (on success).
- LinesPosted (integer count).
- ExceptionCode and ExceptionDetail (on non-success).
- IntakeFormRef (from the transaction).

12.2 Daily Summary Email

At end-of-day the automation emits a summary email to dl-hrops-benefits with subject “[HR-Auto] New-hire onboarding YYYY-MM-DD — {Successes}S / {BE}BE / {AE}AE” and body:

```
Date: 2026-05-10
Polls executed: 72
Items processed: 9
Successes: 8
Business exc.: 1 (DEPENDENTS_MISSING on E1006)
Application: 0
Evidence: BenefitsPostingLog.xlsx
Next dispatcher cycle: 2026-05-10 19:30 (off-hours cadence)
```

12.3 Audit Trail

Each successful transaction leaves three distinct audit artefacts: (1) the row in BenefitsPostingLog.xlsx, (2) the Active enrollment record in BenefitConnect carrying source = api:accrual-onboarding-bot-v2 (BR-12), and (3) the BenefitConnect audit log entry for each API call. In combination these artefacts satisfy the Internal Audit Memo IA-2026-04 evidentiary expectations; OI-06 tracks the ongoing decision on whether per-transaction screenshots are still required.

13. Assumptions, Dependencies, Out-of-Scope

13.1 Assumptions

- The orientation intake process runs to completion before the candidate appears in the Onboarding Queue, i.e. all elections for the hire are already captured on the signed intake form and the dependent/beneficiary records exist in BenefitConnect.
- The internal endpoints listed in §9.2.3 / Appendix A.3 remain operational for partner X-API-Key callers under the tenant's current access policies. If 403 API_NOT_PUBLIC begins to surface, the UI flow (§3.2 / Appendix C) is the documented fallback (OI-07).
- IRS limits used by the consumer are the published 2026 limits and will be updated in place by the Benefits Administration SME prior to the plan-year boundary.
- The secret store is reachable from the automation platform and the secret rotation cadence is compatible with the run schedule.
- Run volume remains below the backpressure ceiling (120 queued items) under all foreseeable conditions.

13.2 Dependencies

- BenefitConnect — availability of /api/public/* (anonymous), /api/employees/me + /api/documents (public-keyed), and the internal endpoints listed in §9.2.3.
- Orientation/Intake source — supply of the elections payload paired with each candidate.
- Automation platform — scheduled trigger, durable queue, REST client capability.
- Secret store — availability at run start.
- SharePoint — availability for the BenefitsPostingLog.xlsx write.
- Email gateway — availability for the daily summary email.

13.3 Explicitly Out of Scope (reiteration)

- Collection of elections (orientation).
- Dependent and beneficiary creation (POST/PATCH/DELETE /api/dependents and PUT /api/beneficiaries are UI-session-only — OI-02, OI-03).
- QLE processing (SCN-11).
- Annual Open Enrollment batch (SCN-08).
- Payroll deduction feed (PI-2024-02).
- Carrier eligibility files (platform-emitted).

14. Automation Suitability Assessment

Suitability is assessed using the Accrual Services Process Engineering criteria. Each dimension is rated High / Medium / Low; the overall rating is qualitative.

Dimension	Rating	Rationale
Volume	HIGH	~180 hires/month sustained, with peaks to 40/day; well above the threshold at which automation ROI is positive.
Frequency	HIGH	Per-event (every queue arrival).
Rule clarity	HIGH	Thirteen deterministic rules (§7), all measurable, with crisp exception codes.
Input stability	HIGH	Onboarding Queue API and candidate schema are stable; the only active source of drift is OI-01.
System accessibility	HIGH	BenefitConnect exposes a REST surface usable with a partner X-API-Key; the public-keyed surface (Appendix A.2) is contractually stable; the internal surface (Appendix A.3) is equivalent in capability to the BenefitConnect UI wizard — it is a workaround to avoid driving the UI — so a UI-automation fallback is feasible if the internal surface ever returns 403 API_NOT_PUBLIC.
Exception-handling fit	HIGH	Exceptions are bounded, catalogued, and have defined owners and SLAs (§10).
Change volatility	MEDIUM	IRS limits update annually; plan catalogue updates at plan-year boundaries; occasional intake-form revisions.
Data sensitivity	MEDIUM	Elections are PII-adjacent; HSA / DCFSA details are more sensitive. Addressed by secret handling (§9.3) and scoped API-key + X-Employee-Id.
Overall	HIGH	Strong candidate; primary residual risk is the reliance on internal endpoints (§9.2.3), mitigated by the run-summary and exception paths in §10 and by the UI-flow fallback documented in §3.2 / Appendix C.

Appendix A — API Reference (Scoped Subset)

This appendix reproduces the subset of the BenefitConnect API Reference relevant to this process, organised by access tier. The full vendor reference is the contractual source-of-truth; see Related Documents.

A.1 Public anonymous (no key required)

Always available, return JSON, used for liveness probes and catalog lookups.

Method	Path	Purpose
GET	/api/public/health	Liveness check. { status, service, version, bootedAt, now }.
GET	/api/public/status	Service status, planYear, openEnrollment window.
GET	/api/public/version	Application + API version.
GET	/api/public/plans	Plan catalogue. ?type=medical dental vision life std ltd fsa hsa commuter retirement.
GET	/api/public/plans/:id	Single plan by id.
GET	/api/public/plan-types	Enum of plan types.
GET	/api/public/carriers	Unique list of carriers.

A.2 Public, X-API-Key required (documented & guaranteed-stable)

Authenticated public-API endpoints. Always available to keyholders and not subject to tenant-side access policies. Both endpoints accept X-Employee-Id (or ?employeeId=) for per-employee scoping.

Method	Path	Purpose
GET	/api/employees/me	Profile of the employee identified by X-Employee-Id (or the API key's default employee).
GET	/api/documents	List of documents available to the employee — plan documents, SBCs, confirmation statements, tax forms, carrier letters.

Example invocation:

```
KEY="accrual-onboarding-bot-v2"
curl -H "X-API-Key: $KEY" -H "X-Employee-Id: E1004" \
  https://accrual-bp-hri.azurewebsites.net/api/employees/me
```

A.3 Internal endpoints (back the BenefitConnect web app; not part of the public API)

Subject to tenant-side access policies; may return 403 API_NOT_PUBLIC at any time. Used here as a programmatic shortcut around the UI; the UI flow in Appendix C is the documented fallback.

Method	Path	Notes
GET	/api/employees	Employee directory; supports filters.
PATCH	/api/employees/me	UI session only.
GET	/api/plans	Internal plan catalog (mirrors /api/public/plans).
GET	/api/plans/:id	Internal single plan.
GET	/api/dependents	Dependents on file (read-only used here).
POST/ PATCH/ DELETE	/api/dependents[:id]	UI session only — see OI-02.
GET	/api/beneficiaries	Beneficiaries (read-only used here).
PUT	/api/beneficiaries	UI session only; sum-to-100 validated — see OI-03.
GET	/api/enrollments/current	Current Active enrollment.
GET	/api/enrollments/history	Prior enrollments.
POST	/api/enrollments	UI session only.
POST	/api/plans/select	Add/replace one election in the draft cart (C-07).
GET	/api/enroll/draft	Read the current draft (C-08).
POST	/api/enroll	Promote draft to a Pending Confirmation enrollment (C-09).
POST	/api/confirm	Confirm pending → Active (alternative C-10; does not resolve the queue item).
GET	/api/onboarding/pending	Cross-employee onboarding queue (D-03).

Method	Path	Notes
GET	/api/onboarding/pending/:id	Single candidate detail.
POST	/api/onboarding/pending/:id/ approve	Confirm pending enrollment AND resolve queue item (preferred C-10).
GET	/api/admin/onboarding-queue	UI session only — HR Admin view.
POST	/api/admin/onboarding- queue/:id/approve	UI session only — equivalent to the Approve button.

Reminder: any call into A.3 may return 403 API_NOT_PUBLIC under tenant-side access policies. The body includes a supportedAlternatives array pointing to A.1 / A.2; the practical fallback for the operator is the UI walkthrough in Appendix C.

Appendix B — Worked Example

Trace of the per-employee transaction in §8.2 — Leilani Tan, the canonical worked example used throughout this PDD. Anchored to the live-tenant record at /admin/onboarding/E1004.

B.1 Candidate snapshot (live-tenant)

Field	Value
BenefitConnect id	E1004
Legal name	Leilani Tan
Email	leilani.tan@accrualbank.example
Phone	+65 8123 5502
Date of birth	Sep 24, 1996
Employee number	ACR-00894
Job title	Associate, Commercial Lending
Department	Commercial Lending — APAC
Division	Accrual Capital
Manager	James Okafor
Work location	Accrual Bank — Singapore HQ
Employee type	Full-Time
Start date	2026-04-26
Status (at trace)	Not Started → Pending Confirmation → Active
Enrollment deadline	2026-05-26

B.2 Consumer trace

EE-tier across medical/dental/vision (no dependents), voluntary life \$100,000, FSA health \$2,200, commuter transit \$150, 401(k) 8% pre-tax. Timings illustrative.

Step	Call	Expected	Timing
C-01	(dequeue)	Item pulled	t+0.0s
C-02	(load secret)	API-key loaded	t+0.1s

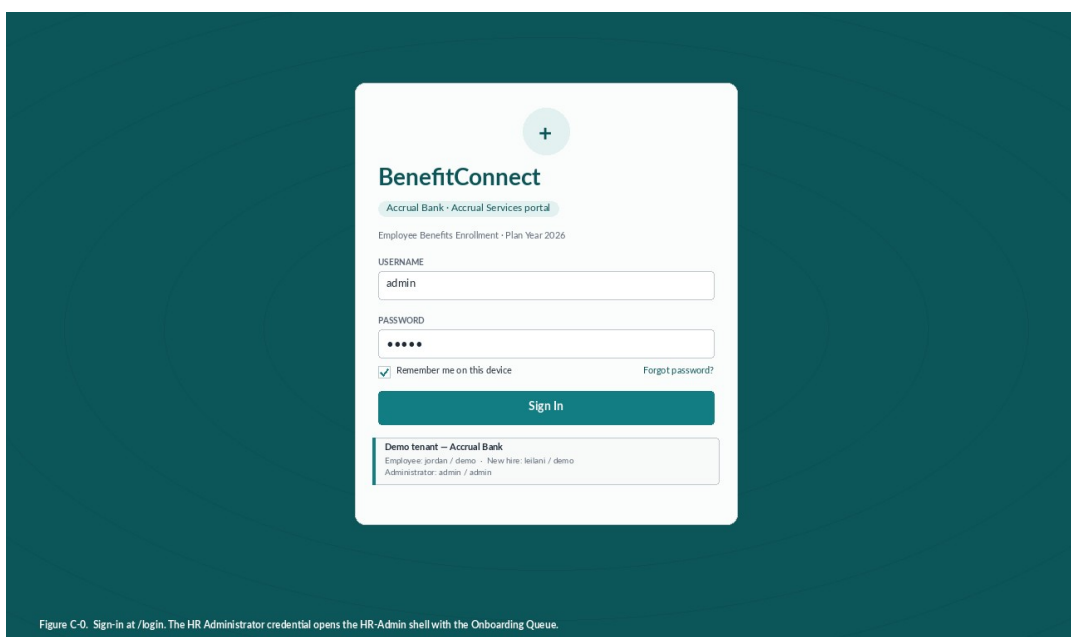
Step	Call	Expected	Timing
C-03	GET /api/employees/me (X-Employee-Id: E1004)	id=E1004	t+0.3s
C-04	GET /api/dependents	0 dependents (EE-only tier)	t+0.5s
C-05	GET /api/beneficiaries	Primary 100% on file	t+0.7s
C-06	GET /api/enrollments/current	null (no prior Active)	t+0.9s
C-07a	POST /api/plans/select (medical)	{ok:true, line:{planId:PLN-MED-PPO,tier:EE}}	t+1.3s
C-07b	POST /api/plans/select (dental)	ok	t+1.6s
C-07c	POST /api/plans/select (vision)	ok	t+1.9s
C-07d	POST /api/plans/select (basic life)	ok	t+2.1s
C-07e	POST /api/plans/select (vol life)	ok, premiumPreview=\$6.50/pp	t+2.4s
C-07f	POST /api/plans/select (STD)	ok	t+2.6s
C-07g	POST /api/plans/select (LTD)	ok	t+2.8s
C-07h	POST /api/plans/select (FSA health)	ok	t+3.0s
C-07i	POST /api/plans/select (commuter)	ok	t+3.2s
C-07j	POST /api/plans/select (401k)	ok, matchPreview included	t+3.4s
C-08	GET /api/enroll/draft	10 lines, totals match	t+3.6s
C-09	POST /api/enroll	enrollmentId=ENR-20260510-0007, status=Pending Confirmation	t+4.0s
C-10	POST /api/onboarding/pending/E1004/approve	status=Active, confirmationNumber=CNF-8392041; queue item resolved	t+4.4s
C-11	(log, resolve queue item)	Log row appended; local queue item Successful	t+4.5s

Appendix C — UI Walkthrough (anchored to the live tenant)

This appendix is the operator's reference for completing benefits onboarding for a single new hire via the BenefitConnect UI. It is also the documented fallback for any transaction the automation declines or escalates. All URLs are relative to <https://accrual-bp-hri.azurewebsites.net>.

C.1 Sign in as HR Administrator

Maggie navigates to `/login`. The sign-in card carries the BenefitConnect logo, the “Accrual Bank · Accrual Services portal” banner, the “Employee Benefits Enrollment · Plan Year 2026” subhead, and the username/password form.



C.2 HR-Admin shell and Onboarding Queue

After sign-in the HR-Admin shell renders the left sidebar — HR OPERATIONS (Dashboard · Onboarding Queue · Onboard New Hire) and MY ACCOUNT (My Profile · Documents) — together with the tenant pill, notifications, help, and identity menu in the top bar.

Maggie clicks Onboarding Queue (URL: `/admin/onboarding`). The page renders a single table titled “Pending HR Onboarding Requests — New hires awaiting benefits enrollment · sorted most-urgent-first”. Columns: Employee · Department · Hire Date · Days Since Hire · Deadline · Status, with a View button on each row. Status values seen in production: Not Started, Draft Saved, Pending Confirmation, Overdue.

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Dashboard

Onboarding Queue

Onboard New Hire

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My Profile
Documents

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HR OPERATIONS · ONBOARDING QUEUE
Onboarding Queue

Accrual Bank ! ? Admin Administrator

Pending HR Onboarding Requests New hires awaiting benefits enrollment - sorted most-urgent-first

EMPLOYEE	DEPARTMENT	HIRE DATE	DAYS SINCE HIRE	DEADLINE	STATUS	
Sebastian Park sebastian.park@accrualbank.example · E1008	Wealth Management	2026-04-05	35	2026-05-05 (30 overdue)	Overdue	View
Mira Gupta mira.gupta@accrualbank.example · E1007	Platform Engineering	2026-04-18	22	2026-05-18 (6d left)	Pending Confirmation	View
Leilani Tan leilani.tan@accrualbank.example · E1004	Commercial Lending — APAC	2026-04-26	14	2026-05-26 (16d left)	Not Started	View
Kim Reyes kim.reyes@accrualbank.example · E1006	Treasury Operations	2026-04-29	11	2026-05-29 (19d left)	Draft Saved	View
Tobi Abiola tobi.abiola@accrualbank.example · E1005	Risk & Compliance	2026-05-05	5	2026-06-04 (25d left)	Not Started	View

Figure C-1. Onboarding Queue at /admin/onboarding — pending candidates sorted most-urgent-first. Leilani Tan (E1004, highlighted) is the worked example.

Live-tenant snapshot at the time of writing: Sebastian Park (E1008, Wealth Management, Overdue), Mira Gupta (E1007, Platform Engineering, Pending Confirmation), Leilani Tan (E1004, Commercial Lending — APAC, Not Started), Kim Reyes (E1006, Treasury Operations, Draft Saved), Tobi Abiola (E1005, Risk & Compliance, Not Started). Maggie selects Leilani's row and clicks View, which opens /admin/onboarding/E1004.

C.3 Candidate detail page

URL: /admin/onboarding/:employeeId — for the worked example /admin/onboarding/E1004. The page heading reads “Onboarding · {Legal Name}” with the candidate's status badge in the upper-right. Three panels render: CANDIDATE (Preferred Name, Email, Phone, Date of Birth, Employee Number), EMPLOYMENT (Job Title, Department, Division, Manager, Work Location, Employee Type, Start Date), and ONBOARDING STATUS (Status, Days Since Hire, Enrollment Deadline, Days Until Deadline). The page footer carries the green Approve button, a Cancel button, and the explanation “Approving will mark this request resolved and (if a pending enrollment exists) confirm it.”

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HR OPERATIONS · ONBOARDING QUEUE

Onboarding · Leilani Tan

Accrual Bank

!

?

AA Admin Administrator

← Back to Onboarding Queue

Leilani Tan

leilani.tan@accrualbank.example · E1004

Not Started

CANDIDATE

Preferred Name	Leilani
Email	leilani.tan@accrualbank.example
Phone	+65 8123 5502
Date of Birth	Sep 24, 1996
Employee Number	ACR-00894

EMPLOYMENT

Job Title	Associate, Commercial Lending
Department	Commercial Lending — APAC
Division	Accrual Capital
Manager	James Okafor
Work Location	Accrual Bank — Singapore HQ
Employee Type	Full-Time
Start Date	2026-04-26

ONBOARDING STATUS

Status	Not Started
Days Since Hire	14 days
Enrollment Deadline	2026-05-26
Days Until Deadline	16 days

✓ Approve

Cancel

Approving will mark this request resolved and (if a pending enrollment exists) confirm it.

Figure C-2. Candidate detail at /admin/onboarding/E1004 — Leilani Tan. The Approve button confirms the pending enrollment and resolves the queue item in a single action.

C.4 Enrollment wizard (per-employee)

If the candidate is in Not Started, Maggie first walks the 6-step BenefitConnect enrollment wizard for the employee. The wizard breadcrumb is visible at the top of every step: 1 Personal Info → 2 Dependents → 3 Medical → 4 Dental / Vision / Life → 5 Savings & 401(k) → 6 Review & Submit. Each step carries a Back / Save draft / Next footer.

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BENEFITS · ENROLLMENT
Benefits Enrollment

Accrual Bank ! ? Leilani Tan

1 Personal Info 2 Dependents 3 Medical 4 Dental / Vision / Life 5 Savings & 401(k) 6 Review & Submit

PERSONAL INFO (read-only — sourced from HRIS)

LEGAL NAME Leilani Tan	PREFERRED NAME Leilani
DATE OF BIRTH Sep 24, 1996	EMAIL leilani.tan@accrualbank.example
PHONE +65 8123 5502	EMPLOYEE NUMBER ACR-00894
JOB TITLE Associate, Commercial Lending	DEPARTMENT Commercial Lending — APAC
DIVISION Accrual Capital	MANAGER James Okafor
WORK LOCATION Accrual Bank — Singapore HQ	START DATE 2026-04-26

← Back Save draft Next →

Figure C-3. Enrollment wizard, Step 1 — Personal Info (read-only, HRIS-sourced).
The breadcrumb at top shows all 6 steps; the operator advances through Dependents, Medical, Dental/Vision/Life, Savings & 401(k), and Review & Submit.

- Step 1 — Personal Info: read-only HRIS-sourced identity. Operator confirms accuracy and clicks Next.
- Step 2 — Dependents: confirms dependents created during orientation. Adding/removing dependents requires the Dependents-management UI (POST/PATCH/DELETE /api/dependents are UI-session-only).
- Step 3 — Medical: select plan and tier (EE / EE+Spouse / EE+Child / Family) from the catalogue dropdown.
- Step 4 — Dental / Vision / Life: dental plan + tier; vision plan + tier; basic life (toggle); voluntary life (face amount, with premium preview). STD / LTD / AD&D toggles.
- Step 5 — Savings & 401(k): FSA health / FSA dependent-care annual; HSA annual (HDHP-only); commuter transit / parking monthly; 401(k) deferral percentages.
- Step 6 — Review & Submit: visual reconciliation of all elected lines; Submit button promotes the candidate from Draft Saved → Pending Confirmation.

C.5 Approve to Active

Maggie returns to /admin/onboarding/:employeeId and clicks Approve. The candidate's status transitions to Active and the request is removed from the queue. Equivalent automation call: POST /api/onboarding/pending/:id/approve (Appendix A.3).

C.6 Onboard New Hire (manual creation)

If the upstream HRIS feed has not yet supplied the new-hire record, the operator can create the candidate manually at /admin/onboarding/new. The form captures Identity (First/Last/Preferred Name, Email, Phone, DOB) and Employment (Start Date, Employee Type, Base Salary, Job

Title, Department, Division, Manager, Work Location). Submitting adds the new hire to the roster and the Onboarding Queue immediately.

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HR OPERATIONS · ONBOARD NEW HIRE

Onboard New Hire

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AA Admin Administrator

← Back to Onboarding Queue

Onboard a New Hire

Add a new employee to the roster and queue them for benefits enrollment.

IDENTITY

FIRST NAME *

LAST NAME *

PREFERRED NAME

Defaults to first name

EMAIL

Auto-generated if blank

PHONE

DATE OF BIRTH

dd/mm/yyyy

EMPLOYMENT

START DATE *

10/05/2026

EMPLOYEE TYPE

Full-Time ?!

BASE SALARY (USD)

JOB TITLE *

DEPARTMENT *

DIVISION

Accrual Services ?!

MANAGER

Full name

WORK LOCATION

Accrual Bank — Singapore HQ

Onboard Employee

Cancel

The new hire is added to the roster and appears in the Onboarding Queue immediately.

Figure C-4. Onboard New Hire form at /admin/onboarding/new — used when HR creates a candidate manually rather than receiving the record from the upstream HRIS feed.

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Appendix D — Glossary

Term	Definition
Active enrollment	An enrollment record in BenefitConnect whose status is Active; the authoritative state for the plan year.
Candidate	A pending onboarding record visible in /admin/onboarding.
Consumer	The automation component that dequeues transaction items and calls BenefitConnect; see §5.4.
Dispatcher	The automation component that polls the Onboarding Queue and emits transaction items; see §5.2.
Draft	Server-side accumulation of enrollment lines materialised by POST /api/plans/select and finalised by POST /api/enroll.
EOI	Evidence of Insurability — out of scope per BR-04.
HDHP	High-Deductible Health Plan; prerequisite for HSA contributions per BR-07.
Onboarding Queue	/admin/onboarding — list of pending candidates.
Partner API-key	X-API-Key issued by BenefitConnect; scoped to accrual-onboarding-bot-v2 for this process.
PDD	Process Definition Document — this document.
QLE	Qualifying Life Event; out of scope (SCN-11).
Transaction item	A single new-hire onboarding task on the NewHireOnboarding queue; see §8.2.
X-Employee-Id	Header that scopes a partner call to a specific employee in the tenant.